

Validation & Verification

Mechatronics & Software

Team 3 - SmartRead: Instant Translation with Wearable Device

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Introduction

In an increasingly interconnected world, the ability to quickly and accurately translate text from one language to another is more important than ever. However, traditional translation methods often involve time-consuming steps, such as taking out a smartphone, opening the translation app, focusing on the text, capturing an image, and then translating. These steps can be inconvenient in fast-paced environments or when handling multiple translations.

SmartRead is an innovative translation smart system designed to boost this process. By integrating advanced hardware and software, SmartRead allows users to capture text instantly with a simple button press. The system automatically captures an image, processes the text, and provides a translation in seconds. Additionally, SmartRead offers a user-friendly interface where users can revisit their saved photos and translation history, making it easier to reference past translations and track their progress.

SmartRead is designed for anyone who needs quick and reliable translations, from travellers and students to professionals and language enthusiasts. By simplifying the translation process and providing a seamless user experience, SmartRead aims to make language barriers a thing of the past.

Purpose

The purpose of this document is to ensure the SmartRead system meets the requirements outlined in the System Requirements & Hazard Analysis document. It provides a clear process for checking that the system works correctly, follows design specifications, and performs reliably in real-world situations. By outlining testing methods, this document helps identify potential problems early, ensuring the system meets safety, security, performance, and functionality expectations. It also helps confirm that the system can handle different conditions without failure. This document ensures that the system is properly tested and works as intended.

Scope

The scope of this document gives the verification and validation process for the SmartRead system by describing how the system will be tested. Through the testing, the system is evaluated to ensure it meets all defined requirements, including safety, security, performance, and functionality. The document describes the methods used to test the system's behaviour under various conditions and verify its compliance with design specifications. This includes developing test cases for both integration and unit testing, ensuring that each requirement is properly

assessed and validated. The document also defines the approach for identifying and addressing any issues that arise during testing, helping to improve the system's reliability and effectiveness.

Constants

Constant	Nominal value / Range
<u>CONST_MAX_TOTAL_WEIGHT</u>	225 grams
CONST_LIGHTING_RANGE	300 ~ 750 lux
<u>CONST_MAX_POWER_CONSUMPTION</u>	1.55 W
CONST_WIFI_SPEED_RANGE	802.11 b/g/n
CONST_MAX_MICROCONTROLLER_DIM	27 * 40.5 * 4.5 (± 0.2) mm
CONST_MIN_RESOLUTION	40 $\times$ 30
CONST_MAX_RESOLUTION	1600 $\times$ 1200
CONST_BATTERY_CAPACITY_RANGE	1000 - 2000 mAh
<u>CONST_BATTERY_DURABILITY_STD</u>	5.8 ~ 10 hours
<u>CONST_BATTERY_DURABILITY_IDLE</u>	5.8 ~ 300 hours
<u>CONST_MAX_OVERALL_TIME</u>	2.8 seconds
<u>CONST_MAX_IMAGE_CAPTURE_TIME</u>	935 ms
<u>CONST_MAX_PROCESSING_TIME</u>	1.4 seconds
<u>CONST_MAX_TRANSMISSION_TIME</u>	32 ~ 436 ms
CONST_MAX_APPLICATION_STORAGE	50 Mb
<u>CONST_MIN_TRANSLATION_ACCURACY</u>	78%

Table 2. Constants

Traceability Matrix

Below is a Traceability Matrix showing the traceability between requirements and test cases. Note that not all test cases are listed here since some tests are constructed based on requirement (i.e. black-box testing), while the others come from code inspection (i.e. white-box testing). Instead, the latter ones are created to test whether the internal modules are working and not included in the traceability matrix.

For a more readable version of this traceability, follow this [link](#) to a Google Sheets file.

Requirement No.	Description	Associated Test Case(s)
Human Safety Requirements		
HSR-1	The system shall use batteries that comply with safety standards to prevent chemical leakage.	IT-10, IT-11, IT-12, IT-15
HSR-2	The system shall prevent overcharging, overheating, or explosions.	IT-15, UT-49, UT-50
HSR-3	The system should ensure safe operation to prevent short circuits.	UT-49, UT-50
HSR-4	The system should ensure proper insulation to prevent electric shocks.	IT-16
HSR-5	The system should protect live components from liquid exposure.	IT-17
HSR-6	The system should control and dissipate heat during operation.	IT-18
HSR-7	The wearable device's surfaces, corners and edges shall be smooth to prevent direct injury to users.	IT-13, IT-14, IT-19
HSR-8	The User Guide should provide instructions on safety inspections and disposal of batteries	IT-20
Environmental Safety Requirements		
ESR-1	The wearable device shall minimize material usage to reduce environmental impact.	IT-21

ESR-2	The wearable device shall use materials that are recyclable and sourced sustainably.	IT-21
ESR-3	The wearable device should lower power consumption during low-usage states.	IT-10, IT-11, IT-12, IT-28, IT-29
ESR-4	The wearable device should be built with materials that comply with environmental safety standards to avoid hazardous substances.	IT-21
ESR-5	The User Guide should label materials to enable safe disposal and recycling.	IT-20
Data Security Requirements		
DSR-1	The system should restrict unauthorized or malicious access.	IT-34 to IT-38, It-44 to IT-51
DSR-2	The system should protect user credentials and secure all transmitted data.	IT-26, IT-27, IT-34 to IT-38, It-44 to IT-51
System & Performance Requirements		
R-1	The system shall support small-volume translation, specifically for text entries containing 50 words or fewer.	IT-01 to IT-05, IT-30 to IT-33
R-2	The system shall display the final translated text within <i>CONST_MAX_OVERALL_TIME</i> seconds starting from receiving a button press input from the user.	IT-01, IT-02
R-3	The system shall be able to identify selected language text from the object, enabling reliable translation with an overall accuracy of at least <i>CONST_MIN_TRANSLATION_ACCURACY</i> under <i>CONST_LIGHTING_RANGE</i> .	IT-04, UT-8 to UT-17
R-4	The system should have a battery that provides a minimum of <i>CONST_BATTERY_DURABILITY_STD</i> hours of active use under standard usage conditions or <i>CONST_BATTERY_DURABILITY_IDLE</i> hours on standby.	IT-10 to IT-12
R-5	The system shall be able to do English-French translation.	IT-30 to IT-32, UT-04

R-6	The wearable device component of the system shall be resizable or elastic to fit most users and not cause users any discomfort.	IT-13
R-7	The system shall inform users about the device's status and promptly notify them of any changes in these conditions in a clear and straightforward manner.	IT-06 to IT-09, IT-22 to IT-25, IT-27
R-8	The system should ensure that the hardware storage utilized for application deployment does not exceed <code>CONST_MAX_APPLICATION_STORAGE</code> for static data, excluding temporary cache memory for images and text.	IT-51
R-9	The wearable device shall weigh no more than <i>CONST_MAX_TOTAL_WEIGHT</i> grams to minimize physical strain.	IT-14
R-10	The system should allow users to save translated phrases for later access and provide a method to view the stored phrases at any time.	UT-31, UT-32, UT-40, UT-41, UT-39 to UT-43
R-11	The system shall recognize text in formal fonts like Arial and Calibri.	UT-18 to UT-21

Table 3. Traceability Matrix

Validation

Validation is a critical process that ensures the defined project requirements lead to a product that meets client expectations. As international students who would benefit from this translation device, we serve as both the designers and part of the target user group. Also, we have been taking feedback from our TA Stefan and Professor Wassying to guide our project. Therefore, we can believe that our defined requirements align with the needs of the intended users. This document will focus more on the verification process.

There are tests (IT-13, IT-26, IT-31, UT-40) that can demonstrate the device behavior meets the desired goal.

Test Coverage Criteria

Integration Testing

The primary goal of integration testing is to ensure that SmartRead not only provides users with timely and accurate translations when the device is used in an expected manner but also handles unforeseen circumstances safely and intuitively. To assure stakeholders of such quality, the integration test cases need to achieve sufficient coverage for both requirements and the **input domain**. While the coverage for requirements is demonstrated through the traceability matrix, input domain coverage is maximized by adhering to a set of predefined rules.

1. Integration tests should verify that all monitored variables are appropriately converted into controlled variables while also covering the interactions between subsystems. This approach emphasizes the principle of separation of concerns, isolating the requirements for each subsystem and consolidating their implementations.
2. The input domain is divided into three categories: standard input, boundary input, and erroneous input. All applicable test scenarios are evaluated using at least one input from each category to ensure the system and all its subsystems can handle inputs from various ranges effectively.

By following the guidelines above, integration testing shall be able to thoroughly examine the behaviour of SmartRead and its components under diverse circumstances. The reliability of the system is thus ensured, providing confidence in its performance, accuracy, and robustness across various real-world scenarios.

Unit Testing

Unit tests are constructed using the requirements and code inspection, but there will be an emphasis on unit tests' code coverage as the majority of requirements have been verified using integration tests. There are a few options for unit testing coverage criteria: statement coverage, branch coverage, conditional coverage, combinatorial coverage, path coverage, and modified condition/decision coverage.

Statement coverage is the easiest to achieve. However, achieving a perfect statement coverage provides little evidence of system robustness and does not guarantee the system handles various inputs appropriately. Conditional coverage, path coverage, and combinatorial coverage are highly effective in identifying implementation flaws, but it becomes extremely impractical to achieve high coverage under these criteria considering the large codebase and the project's time constraints. While MC/DC coverage offers a good balance between cost and performance, it is not selected due to the complexity of constructing such test cases. Considering the factors above, SmartRead has chosen **decision coverage** as the coverage criteria. Since the program's flow is partially verified through integration testing, decision coverage ensures that each decision branch is tested, helping to identify and eliminate potential issues at a reasonable cost. The [Unit Testing](#) section of this document will provide a detailed list of unit tests that were constructed to achieve adequate decision coverage.

API Testing

API invocations are tested with both integration and unit testing. Integration testing involves utilizing the actual API and interacting with it by providing meaningful inputs. Unit tests typically mock API calls to speed up test runs, prevent hitting API rate limits, and ensure more predictable test behaviours.

Verification Tests

Integration Testing

The standard of pass/fail is that if the actual output aligns with the expected output, the test passes. Otherwise, the test fails. ‘/’ means not applicable or the test description is trivial.

Test ID	Test Name	Description	Input	Expected Output	Actual Output	Pass /Fail
General System Integration						
IT-01	30WordsProcessingTimeTest	Ensure users can observe the translation results within a certain time period after they capture an image	Button press Image with ~30 words	Translation displayed within 2.8 seconds	Translation displayed After 19 seconds	Fail
IT-02	70WordsProcessingTimeTest	Same as IT-01	Button press Image with ~70 words	Translation displayed within 3.5 seconds	Translation displayed After 25 seconds	Fail
Both failed due to the time limit being exceeded.						
IT-03	Multi-DeviceProcessingTest	If multiple devices (wearable and UI) are running in parallel, verify whether the correct images are displayed on the UI	Button press on both devices (Wearable_A captures Image_A Wearable_B captured Image_B)	Image_A displayed on UI_A Image_B displayed on UI_B	Image_A displayed on UI_A Image_B displayed on UI_B	Pass
IT-04 ¹	DifferentImageConditionHandlingTest	/	/	/	/	/
IT-05	NoTextImageProcessingTest	Verify if the UI displays a corresponding message when there is no text in the captured image	Button press Image with no text	Warning message displayed on the UI	/	Fail

¹ IT-04 consists of a total of 5 unit tests covering various distance/lighting conditions. The detailed specifications are provided in Appendix A.

IT-05 produced inconsistent results due to the "black-boxed" machine learning model. It is identified as failed because the segmentation model identifies one segment for most of the "text-free" images.						
IT-06	BatteryLifeWarningTest	Verify low battery warning message for users	Low battery power (less than 3.2V)	A low battery notification is shown on the UI	A low battery notification is shown on the UI	Pass
IT-07	TemperatureUnderWarningTest	Verify low-temperature warning of electronic components for users	Device Temperature below the minimum operating temperature	UI displays a warning indicating device temperature is too low	UI displays a warning indicating device temperature is too low	Pass
IT-08	TemperatureOverWarningTest	Verify high-temperature warning of electronic components for users	Device temperature above the maximum operating temperature	UI displays a warning indicating device temperature is too high	UI displays a warning indicating device temperature is too high	Pass
IT-09	ConnectionLostWarningTest	Verify WiFi connect lost warning for users	Cut WiFi connection	Navigate to the connection guide page	Navigate to the connection guide page	Pass
IT-10	StandardBatteryDurabilityTest	Verify battery standard durable time	With a fully-charged battery, the device runs in a standard manner	Battery lasts about 23h (calculation see Appendix A 1)	No available techniques to test this.	/
IT-11	LightSleepBatteryDurabilityTest	Verify battery life in light sleep mode	With a fully charged battery, the device runs in light sleep mode	Battery lasts about 89h (calculation see Appendix A 2)	No available techniques to test this.	/
IT-12	ExtremeBatteryDurabilityTest	Verify battery life in prolonged active use	With a fully charged battery, the device runs in active mode	Battery lasts about 6h (calculation see Appendix A 3)	No available techniques to test this.	/
IT-13	WearableDeviceComfortTest	Verify the wearable device is comfortable for various users to wear.	Let various users wear the device for 30 minutes.	Users feel the device is suitable and comfortable.	Users feel the device is suitable and comfortable.	Pass
IT-14	WearableDeviceWeightTest	Verify the overall system weight would not exceed the threshold	Assembled wearable device	The weight of the device is less than 225 grams	The device weighs 112g.	Pass
IT-15	BatteryComplianceTest	Verify the battery meets safety standards and does not leak	Battery exposed to 60 °C for 4 hours.	No chemical leakage, swelling, or	No available techniques to test this.	/

				deformation		
IT-16	InsulationTest	Ensure proper insulation to prevent electric shocks	The headband is printed in PLA material, which is non-conductive; the voltage of the microcontroller is 5V only.	The headband is electrically insulating.	The headband is electrically insulating.	Pass
IT-17	WaterIngressTest	Verify water does not enter through button holes, charging ports, or seams, affecting functionality	Simulate sweat exposure (high humidity test) and water splashes on buttons and changing ports	System continues functionality properly, buttons remain responsive, charging still works, no internal condensation	System continues functionality properly, buttons remain responsive, charging still works, no internal condensation	Pass
IT-18	Thermal ManagementTest	Verify the system maintains safe operating temperatures under load.	Run the system for 1 hour under maximum power consumption, then measure internal temperature and environment temperature	Internal temperature remains within safe limits, and there is no excessive heat buildup compared to the environment	Internal temperature remains within safe limits, and there is no excessive heat buildup compared to the environment	Pass
IT-19	PhysicalSafetyTest	Ensure device edges are smooth and safe	Inspect and test edges with a finger probe	No sharp edges, no injury risk	No sharp edges, no injury risk	Pass
IT-20	SafetyInstruction CheckTest	Verify draft safety checklist exists for battery handling and disposal	Generate a user manual to give instructions on battery handling and disposal.	User manual gives instructions on battery handling and disposal.	User manual gives instructions on battery handling and disposal.	Pass
IT-21	Environmental ComplianceTest	Ensure the device minimizes material usage, uses sustainable/recyclable materials, and complies with environmental safety standards	Review bill of materials, conduct material weight measurement, and check compliance against sustainability certifications	The device uses minimal material, including recyclable and sustainably sourced components, and contains no hazardous substances exceeding regulatory limits	The device uses minimal material, including recyclable and sustainably sourced components, and contains no hazardous substances exceeding regulatory limits	Pass

Electronics & UI Integration						
IT-22	ValidWiFi ConnectionTest	Verify valid WiFi credentials can be connected properly	WiFi credentials (WiFi name and corresponding password)	Users can see the WiFi connection successful window on the UI	Users can see the WiFi connection successful window on the UI	Pass
IT-23	InvalidWiFi ConnectionTest	Verify invalid WiFi credentials can not be connected	WiFi credentials (WiFi name and an invalid password)	Users can see WiFi connection failure on the UI	Users can see WiFi connection failure on the UI	Pass
IT-24	RecoverWiFi ConnectionTest	Verify WiFi reconnection after reconnecting power source	Power source	WiFi can be reconnected without users input	WiFi can be reconnected without users input	Pass
IT-25	APMode ConnectionTest	Verify AP WiFi connection when the system is in use	AP WiFi credentials (AP WiFi name and corresponding password)	AP mode started. Users can enter AP WiFi credentials and connect to it on their phone	AP mode started. User can connect AP WiFi on their phone	Pass
Electronics & Server Integration						
IT-26	ImageCaptureTest	Verify the image can be captured successfully after button (GPIO-14) is pressed	Push button (GPIO-14)	Image was captured successfully and received on server	Image was captured successfully and received on server	Pass
IT-27	SendMessageTo ServerTest	Verify image and system status including WiFi connection, temperature and battery warning can be sent to the server properly	Triggered periodically by system status message and when there's a captured image	Status received on server	Status received on server	Pass
IT-28	LightSleepMode Test	Verify the system is in light sleep mode after the system is not used for a certain period of time	Triggered by the difference between current time and the last time user pressed the push button to capture image	Significant current drop compared to active mode.	In light sleep mode, the system only consumes 14mA, much lower than active mode.	Pass
IT-29	LightSleepMode WakeTest	Verify the system will wake up from light sleep mode upon	Push button (GPIO-14)	Significantly current rise compared to sleep	In active mode, the system consumes	Pass

		user presses the button (GPIO-14)		mode. Users are able to take images upon pressing the button for the second time.	250mA.	
Server & API/Database Integration						
IT-30	30WordsEnFr TranslationTest	Ensure system behaviour when processing a translation request of approximately 30 words.	Random, meaningful texts(str) ~30 words.	Translation results with expected phrases See detailed outputs in Appendix B	Translation results with expected phrases See detailed outputs in Appendix B	Pass
IT-31	50WordsEnFr TranslationTest	Ensure system behaviour when processing a translation request of approximately 50 words.	Random, meaningful texts(str) about 50 words.			Pass
IT-32	70WordsEnFr TranslationTest	Ensure system behaviour when processing a translation request of approximately 70 words.	Random, meaningful texts(str) about 70 words.			Pass
IT-33 ²	LanguagePair TranslationTest	Ensure that translations work for different supported language pairs (English, Chinese, French, Japanese, Italian)	List of random, meaningful texts(str) with 30 words in different supported languages			Pass
UI & API Integration						
IT-34	LogInWithEmpty EmailTest	Verify behavior when logging with empty email	Click on the "Login" button while leaving the email empty	Non-empty email popup displayed	Non-empty email popup displayed	Pass
IT-35	LogInWithRandom StringTest	Verify behavior when logging with random string	"aaa" Click on the "Login" button	Invalid email format popup displayed	Invalid email format popup displayed	Pass
IT-36	LogInWithUnexist EmailTest	Verify behavior when logging with invalid email	" aaa@gmail.com " Click on the "Login"	Email not found popup displayed	Email not found popup displayed	Pass

² UT-07 consists of a total of 19 unit tests covering various language pairs. The detailed specifications are provided in Appendix A.

			button			
IT-37	LogInWithWrongPasswordTest	Verify behavior when logging with incorrect password	Valid email and wrong password Click on the "Login" button	Invalid password popup displayed	Invalid password popup displayed	Pass
IT-38	LogInSuccessfulTest	Verify behavior when doing valid logging in	Valid email and password Click on the "Login" button	Navigate to the home page	Navigate to the home page	Pass
UI & Server & API/Database Integration						
IT-39	FetchUnexistLanguagePreferenceTest	Verify if no warning is displayed when the user has no language preference history	Successful login	Unselected language preferences	Unselected language preferences	Pass
IT-40	FetchLanguagePreferenceWhenServerDownTest	Verify if the UI responds properly when fetching language preferences while the server is down	Successful login Unavailable server	Try again later popup displayed	Network request failed	Fail
IT-41	FetchLangPreferenceSuccessfulTest	/	Successful login	Correct source and target languages	Correct source and target languages	Pass
IT-42	SaveSourceLangWithoutTargetLangTest	Verify if a user can select a source language without specifying a target language	Source language → English No target language selection	Target language required popup displayed	Target language required popup displayed	Pass
IT-43	SaveLanguagePreferenceTest	/	Source language → English Target language → Chinese	The "Save" button changed to an "Update" button	The "Save" button changed to an "Update" button	Pass
IT-44	SignupWithEmptyUsername	/	"" " aaa@gmail.com "	Username required popup displayed	Username required popup displayed	Pass

	Test		"111111" ³			
IT-45	SignupWith EmptyEmail Test	/	"HungryPanda" "" "111111"	Email required popup displayed	Email required popup displayed	Pass
IT-46	SignupWith EmptyPassword Test	/	"HungryPanda" " aaa@gmail.com " ""	Password required popup displayed	Password required popup displayed	Pass
IT-47	SignupWith ExistEmailTest	/	"HungryPanda" An email has been registered "111111"	Email already in use popup displayed	Email already in use popup displayed	Pass
IT-48	SignupWithInvalid EmailFormatTest	/	"HungryPanda" " aaa@gmail " "111111"	Invalid email format popup displayed	Invalid email format popup displayed	Pass
IT-49	SignupWith ShortPasswordTest	/	"HungryPanda" " aaa@gmail.com " "111"	Weak password popup displayed	Weak password popup displayed	Pass
IT-50	SignupSuccessful Test	/	"HungryPanda" " aaa@gmail.com " "111111"	Navigate to the home screen New document created in Firebase	Navigate to the home screen New document created in Firebase	Pass
IT-51	UIApplication StorageSpaceTest	Verify if the installed UI application occupies more than 50 Mb of storage space	App installation	Installed app requires less than 50 Mb	Installed app requires 78 Mb	Fail

Table 4. Integration Tests

³ From top to bottom: username, email, and password. Same for all signup tests.

Unit Testing

Test ID	Test Name	Description	Input	Expected Output	Actual Output	Pass /Fail
TranslationModule						
UT-01	Translator InitializationTest	Ensure the system can successfully initialize a translator instance	Initialization call	Initialization Succeeded	Initialization Succeeded	Pass
UT-02	InvalidTranslator HandlersTest	Verify the translator instance cannot be initialized if module import is failed	Initialization call while simulating module import failure	Initialization Failed	Initialization Failed	Pass
UT-03	NoAvailable TranslatorHandlers Test	Verify the translator instance cannot be initialized if no handlers are specified	Initialization call with empty translator handlers list	Initialization Failed	Initialization Succeeded	Fail
UT-03 failed because the initialization process does not handle the condition that the available translators list is empty.						
UT-04	TranslateTest	Ensure the system can successfully apply the translation function	Random, meaningful texts(str)	Specified Translation result	Specified Translation result	Pass
TextExtractionModule						
UT-05	ExtractorsAnd Segmenters InitializationTest	Ensure the system can successfully initialize extractor and segmenter instances	Initialization call	Initialization Succeeded	Initialization Succeeded	Pass
UT-06	InvalidExtractors And Segmentors HandlersTest	Verify the extractor and segmenter instance cannot be initialized if module import is failed	Initialization call while simulating module import failure	Initialization Failed	Initialization Failed	Pass
UT-07	NoAvailable Extractors AndSegmentors HandlersTest	Verify the extractor and segmenter instance cannot be initialized if no handlers are specified	Initialization call with empty translator handlers list	Initialization Failed	Initialization Succeeded	Fail
UT-07 failed because the initialization process does not handle the condition that the available extractors and segmentors list is empty.						
UT-08	SingleRegion	Ensure the system correctly	Image with one block of	Text class coordinate.	Coordinates in	Pass

	SegmentationTest	identifies and segments a single region of text in an image.	text (with tolerance of other non-text content)	(with tolerant of other non-text content)	Appendix B - Test Inputs & Expected Output	
UT-09	MultipleRegion SegmentationTest	Ensure the system correctly identifies and segments multiple regions of text in an image.	Image with 2+ blocks of text (with tolerance of other non-text content)	All text class coordinates. (with tolerant of other non-text content)		Fail
UT-06 failed due to line spacing; the text composition may cause more sub-images, such as segmentation line by line, but overall, it is still accurate.						
UT-10	NoText SegmentationTest	Confirm that the system does not falsely detect text when none is present in an image.	Image with no block of text (with tolerance of other non-text content)	Empty text class coordinates(list). (with tolerant of other non-text content)	Coordinates in Unit Test Inputs & Expected Output	Pass
UT-11 -1	NormalText ExtractionTest	Ensure accurate text extraction under standard conditions.	Image with text class confirmed.	Extracted text	Extracted text in Appendix B - Test Inputs & Expected Output	Pass
UT-11 -2	NormalText ExtractionTest MultipleTextRegion	Ensure accurate text extraction under standard conditions with multiple text regions.	Image with text class confirmed.	Extracted text		Pass
UT-12	NoTextExtraction Test	Ensure the OCR system does not return false positives when processing an image without text.	Image without text class	Empty string		Pass
UT-13	DarkText ExtractionAccuracy Test	Evaluate OCR accuracy on bright images with reduced visibility.	Image with approximately 300 lux brightness	Extracted text		Pass
UT-14	BrightText ExtractionAccuracy Test	Evaluate OCR accuracy on dark images with reduced visibility.	Image with approximately 750 lux brightness	Extracted text		Pass
UT-15	NormalLighting TextExtraction AccuracyTest	Assess OCR performance when lighting within the system's recommended range.	Image with approximately 500 lux brightness	Extracted text		Pass
UT-16	UnderLighting TextExtraction	Assess OCR performance when lighting exceeds the system's	Image with approximately 100 lux	Extracted text		Pass

	AccuracyTest	recommended range.	brightness			
UT-17	OverLighting TextExtraction AccuracyTest	Assess OCR performance with lighting conditions below the system's recommended range.	Image with approximately 850 lux brightness	Extracted text		Fail
UT-17 failed with accuracy = $0.9362 < 0.95$ (R-3); performance with Overlighting conditions will be considered as part of refinements.						
UT-18	ArialSegmentation Test	Ensure the system can correctly detect and segment text written in Arial font.	Image with block(s) of text in the Arial font (with the tolerance of other non-text content).	All text class coordinates. (with tolerance of other non-text content)	Coordinates in Appendix B - Test Inputs & Expected Output	Pass
UT-19	Calibri SegmentationTest	Ensure the system can correctly detect and segment text written in Calibri font.	Image with block(s) of text in the Calibri font (with the tolerance of other non-text content).	All text class coordinates. (with tolerance of other non-text content)		Pass
UT-20	ArialTextExtraction Test	Ensure accurate OCR performance for text written in Arial.	Segmented Image with Arial text class confirmed.	Extracted text	Extracted text in Unit Test Inputs & Expected Output	Fail
UT-20 failed with accuracy = $0.4667 < 0.95$ (R-3); Such low accuracy was caused by incorrect extracting text line order.						
UT-21	CalibriTextExtraction Test	Ensure accurate OCR performance for text written in Calibri.	Segmented Image with Calibri text class confirmed.	Extracted text	Extracted text in Unit Test Inputs & Expected Output	Pass
TaskManagerModule						
UT-22	QueueInitialization Test	Ensure the system can successfully initialize a queue instance for task execution	Initialization call	Initialization Succeeded	Initialization Succeeded	Pass
UT-23	QueueRemove TaskTest	Ensure a task can be removed from the queue	Attempt to remove a valid task from the queue	Successfully Removed	Successfully Removed	Pass
UT-24	InvalidTaskID RemoveTest	Ensure that attempting to remove a task with an invalid task ID from the queue is handled correctly	Attempt to remove an undefined task ID from the queue	Removed Failed with an appropriate error message	Removed Failed with an appropriate error message	Pass

UT-25	EmptyQueue RemoveTaskTest	Ensure that attempting to remove a task from an empty queue is handled correctly	Attempt to remove a task from an empty queue	Removed Failed with an appropriate error message	Removed Failed with an appropriate error message	Pass
UT-26	QueueAddTask Test	Ensure a task can be added to the queue in the correct order	Add a task in the queue	Added Successfully at the end of the queue	Added Successfully at the end of the queue	Pass
UT-27	DuplicateTask Manager InitializationTest	Ensure the TaskManager instance follows the Singleton design pattern	Attempt to create multiple instances of TaskManager	Only one instance is created	Only one instance is created	Pass
UT-28	TaskExecution SuccessfulTest	Ensure a task in the queue can be executed successfully	Add a valid task to the queue and trigger execution	Executed Successfully	Executed Successfully	Pass
UT-29	TaskExecution ErrorTest	Verify the task execution errors are handled properly	Execute a task known to raise an exception	The error is captured with an appropriate error message	The error is captured with an appropriate error message	Pass
UT-30	UninitializedTask ExecutionTest	Verify the uninitialized task will not be executed	Add an uninitialized task to the queue and trigger execution	The task is not executed and removed the queue	The task is not executed, and removed the queue	Pass
UT-31	SaveTranslation SuccessTest	Ensure that translation history can be saved with the request.	Valid translation result	Translation result saved to mock database	Translation result saved to mock database	Pass
UT-32	SaveTranslation FailedTest	Ensure the method call fails when the translation result is empty	Empty translation result	Method returns false	Method returns false	Pass
UserManagerModule (Routers)						
UT-33	SignUpWithUid AndUserNameTest	Ensure that a new user can successfully create an account.	{"uid": "test_uid", "username": "test_user"}	200, "User signed up successfully"	200, "User signed up successfully"	Pass
UT-34	SignUpWithoutUid Test	Ensure that a new user cannot successfully create an account without Uid.	{"username": "test_user"}	400, "Missing user ID"	400, "Missing user ID"	Pass
UT-35	SignUpWith	Ensure that a new user cannot	{"uid": "test_uid",	400, "User ID already	400, "User ID already	Pass

	DuplicateUserTest	successfully create an account with a duplicate user ID	"username": "test_user"}	exists"	exists"	
UT-36	SaveLangPreferenceWithUidTest	Ensure that users can successfully set the source language and target language.	{"uid": "test_uid", "sourceLang": "French", "targetLang": "Chinese"}	200, "Language preference saved successfully"	200, "Language preference saved successfully"	Pass
UT-37	SaveLangPreferenceWithoutUidTest	Ensure that users cannot successfully set source language and target language without Uid.	{"uid": "", "sourceLang": "French", "targetLang": "Chinese"}	400, "Missing user ID"	400, "Missing user ID"	Pass
UT-38	GetLangPreferenceWithUidTest	Ensure that their preferred language settings can be selected and saved.	Get request to /lang-pref?uid=test_uid	200, {"source-lang": "French", "target-lang": "Chinese"}	200, {"source-lang": "French", "target-lang": "Chinese"}	Pass
UT-39	GetLangPreferenceWithoutUidTest	Ensure that their preferred language settings cannot be selected and saved without Uid.	Get request to /lang-pref	400, "Missing user ID"	400, "Missing user ID"	Pass
UT-40	FetchTranslationHistoryTest	Ensure that translation history can be fetched with the request.	Get request to /history?uid=test_uid	Mocked Firebase response	Mocked Firebase response	Pass
UT-41	FetchEmptyTranslationHistoryTest	Ensure that empty translation history can be fetched with the request.	Get request to /history?uid=test_uid	[]	[]	Pass
AuthenticationModule (UI)						
UT-42	LogInSuccessfulTest	Ensure a user can successfully log in to the UI software with a valid email and password combination	"test@example.com", "password123"	{status: 1, message: "Log in successful!"}	{status: 1, message: "Log in successful!"}	Pass
UT-43	InvalidCredentialsLogInTest	Ensure a login request fails while the password specified is incorrect (or email unregistered/empty credentials)	"invalid-email", "password123"	{status: 0, message: "Invalid email format."}	{status: 0, message: "Invalid email format."}	Pass
UT-44	InvalidEmailFormatTest	Ensure a login request with an invalid email format fails	"test@example.com", "wrongpassword"	{status: 0, message: "Email not found or	{status: 0, message: "Email not found or	Pass

				invalid password."}	invalid password."}	
UT-45	LoginNetworkErrorTest	Ensure correct error message is returned when a login request fails due to an unexpected error	Mocked network error	{status: 0, message: "Error: Network error."}	{status: 0, message: "Error: Network error."}	Pass
All unit tests for AuthenticationModule in UI pass due to matching responses from the Firebase API.						
ElectronicsModule						
UT-46	TemperatureSensorTest	Verify that the temperature sensor is able to measure device's CPU temperature.	Put the device at room temperature.	The temperature reading should be around 45 degrees Celsius.	The temperature reading fluctuates a lot between 150-200, not reliable.	Fail
UT-47	PowerOnSwitchTest	Verify that the power switch works	Turn the power switch on and off.	When the power switch is on, the power booster is enabled and the blue LED turns on. Vice versa.	When the switch is on, power is on and the blue LED is on. When the switch is off, the power is off and the blue LED is off.	Pass
UT-48	PowerBoosterTest	Verify that the power booster can boost 3.7V battery power over 5V.	3.7V battery	Power booster output to be over 5V	Power booster outputs 5.2V	Pass
Passed according to the multimeter's reading						
UT-49	OverChargeProtectionTest	Ensure the system prevents overcharging and overheating	Charge battery beyond rated voltage	Charging stops, no overheating	Charging stops, no overheating	Pass
UT-50	ShortCircuitProtectionTest	Ensure the system detects and prevents short circuits without damage	Measure voltage across power rails to simulate a short circuit	Voltage drops to 0V or stays within safe limits.	Voltage drops to 0V or stays within safe limits.	Pass

Table 5. Unit Tests

Appendix A - Battery Life Calculations

1. Battery life under standard usage calculation:
 - a. Background info:
 - i. Battery 2100mAh
 - ii. ESP32-CAM light sleep mode consumption 20mA
 - iii. ESP32-CAM active mode consumption 300mA
 - b. Assume power efficiency of 85% (for power lost during power boost and circuit)
 - c. Assume standard usage is 80% light sleep mode and 20% active mode
 - d. Calculation:
 - i. Average current: $20\text{mA} \times 80\% + 300\text{mA} \times 20\% = 76\text{mA}$
 - ii. Effective capacity: $2100\text{mAh} \times 85\% = 1785\text{mAh}$
 - iii. Battery life: $1785\text{mAh} / 76\text{mA} = 23.48\text{h}$
2. Battery life in light sleep mode calculation:
 - a. Background and power efficiency assumptions are the same as above.
 - b. Calculation: $1785\text{mAh} / 20\text{mA} = 89.25\text{h}$
3. Battery life in active mode calculation:
 - a. Background and power efficiency assumption are the same as above.
 - b. Calculation: $1785\text{mAh} / 300\text{mA} = 5.95\text{h}$

Appendix B - Test Inputs & Expected Output

IT-30

Input: The sun rises in the east, and birds begin to sing as the morning light spreads. Farmers wake up early to tend their crops and prepare for another day of work.

Expected phrases in the output: [

"soleil se lève",
"oiseaux commencent à chanter",
"agriculteurs se lèvent tôt",

]

IT-31

Input: Technology has transformed the way people communicate and work. With the rise of the internet, instant messaging and video conferencing have become essential. Businesses now rely on cloud computing, artificial intelligence, and automation to improve efficiency and stay competitive in an ever-changing digital world.

Expected phrases in the output: [

"messagerie instantanée",
"intelligence artificielle",
"monde numérique",

]

IT-32

Input: In the modern era, technology shapes the way people interact, work, and learn. The evolution of smartphones, social media, and artificial intelligence has created an interconnected world. Businesses leverage cloud computing and data analytics to optimize decision-making. Automated systems enhance efficiency, reducing human workload. As innovation continues, the role of digital transformation becomes increasingly crucial in every industry.

Expected phrases in the output:[

"ère moderne",
"technologie",
"intelligence artificielle",
"cloud computing",
"automatisés",
"innovation",

]

IT-33

English → Chinese

Input: The moon shines brightly at night, illuminating the silent streets. People rest in their homes, preparing for the challenges of the next day.

Expected phrases in the output: ["月", "明亮", "照亮", "街道", "人们", "休息", "挑战"]

English → Japanese

Input: The mountains stand tall in the distance, covered in a blanket of snow. Travelers admire the breathtaking view, feeling at peace with nature.

Expected phrases in the output: ["山", "立", "雪に覆われ", "旅行者", "景色", "自然"]

English → Italian

Input: A warm breeze flows through the city, bringing the scent of blooming flowers. The streets are lively with people enjoying the sunshine and music.

Expected phrases in the output: [

"brezza calda",

"fiori",

"persone",

"godono",

"musica",

]

Chinese → English

Input: 太阳在早晨的天空升起，温暖的阳光洒满大地。农民们开始他们新的一天，而鸟儿在树枝上歌唱。

Expected phrases in the output: ["sun rises", "warm", "farmers", "birds", "sing"]

Chinese → French

Input: 太阳升起, 照亮了沉睡的城市。鸟儿在枝头歌唱, 人们开始忙碌的一天。

Expected phrases in the output: [

"soleil se lève",
"oiseaux chantent",
"les gens commencent",
"journée",

]

Chinese → Japanese

Input: 太阳升起, 温暖的大地。鸟儿歌唱, 人们开始他们的工作。

Expected phrases in the output: ["太陽が昇", "鳥", "歌", "人々", "仕事"]

Chinese → Italian

Input: 清晨, 城市逐渐苏醒, 街道上开始热闹起来。商贩摆出新鲜蔬果, 人们陆续出门上班。

Expected phrases in the output: [

"la città si",
"frutta e verdura fresca",
"lavoro",

]

French → English

Input: Le vent souffle doucement, portant les parfums des fleurs en pleine floraison. Les rues sont pleines de vie alors que les gens profitent du soleil.

Expected phrases in the output: [

"wind blows",
"scent",
"flowers",
"streets",
"people enjoy the sun",

]

French → Chinese

Input: Le soleil se lève à l'horizon, illuminant progressivement la ville endormie. Les oiseaux chantent et les gens se préparent pour une nouvelle journée.

Expected phrases in the output: ["太阳", "升起", "城市", "照亮", "鸟", "歌唱"]

French → Japanese

Input: Les collines verdoyantes s'étendent à perte de vue, accueillant la brise matinale. Les fermiers nourrissent le bétail alors que la journée commence.

Expected phrases in the output: ["緑", "丘", "朝", "風", "農家", "家畜", "餌"]

French → Italian

Input: Le marché du village s'anime dès l'aube, offrant des produits frais et variés. Les habitants discutent joyeusement tout en faisant leurs achats.

Expected phrases in the output: ["mercato", "prodotti freschi", "allegrement", "shopping"]

Japanese → English

Input: 夜明けとともに海面が輝き、漁師たちは船を出す準備を始める。浜辺では子供たちが貝を集めて楽しんでいる。

Expected phrases in the output: [

"dawn",

"sea",

"fishermen",

"children",

"collect",

"shell",

]

Japanese → Chinese

Input: 太陽が昇ると、街の通りは活気に満ち溢れる。人々は新しい一日の始まりに期待を抱いている。

Expected phrases in the output: ["太阳", "升", "街道", "人们", "开始"]

Japanese → French

Input: 美しい庭園には色とりどりの花が咲き、穏やかな噴水の音が響いている。観光客は写真を撮りながら散策を楽しむ。

Expected phrases in the output: [

"jardin",
"magnifique",
"fleurs colorées",
"touristes",
"prendre des photos",

]

Japanese → Italian

Input: 遠くに見える山々は朝靄に包まれ、静けさの中に神秘的な雰囲気醸し出す。ハイカーたちは新鮮な空気を吸いながら山道を歩き始める。

Expected phrases in the output: [

"montagne",
"lontananza",
"avvolte",
"escursionisti iniziano a camminare",

]

Italian → English

Input: Una leggera brezza soffia attraverso i campi, portando il profumo della primavera. Gli agricoltori lavorano diligentemente sotto il cielo azzurro.

Expected phrases in the output: [

"breeze blows",
"fields",
"scent of spring",
"farmers",
"work",
"diligently",

]

Italian → Chinese

Input: All'alba, le luci della città cominciano a spegnersi mentre il mercato apre i battenti. Le persone acquistano prodotti freschi per la giornata.

Expected phrases in the output: ["城市的灯光", "开始", "市场", "人们购买", "新鲜"]

Italian → French

Input: Verso sera, le strade si tingono di luci e ombre, mentre la gente torna a casa. Un senso di tranquillità cala sulla città, promettendo una notte serena.

Expected phrases in the output: [

"夕方",
"影",
"人々",
"帰",
"街",
"穏やかな夜",

]

Italian → Japanese

Input: Nel pomeriggio, una brezza leggera attraversa la piazza, portando il profumo di caffè. I turisti si fermano per godersi il momento al tavolo di un bar.

Expected phrases in the output: [

"après-midi",
"traverse la place",
"café",
"touristes",
"du moment",

]

UT-08 / UT-11-1

Input:

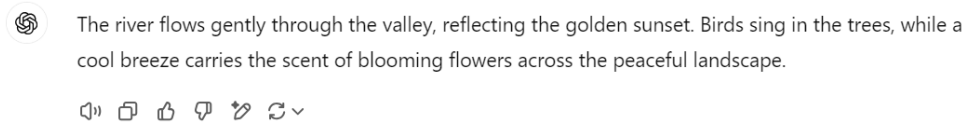


Figure 1. Single Region Text Image

Expected segmentation output (left, right, top, bottom):

[(43, 609, 28, 62)]

- Considered pass if each coordinate is within 5 pixels of the expected value

Expected text extraction output:

"The river flows gently through the valley, reflecting the golden sunset. Birds sing in the trees, while a cool breeze carries the scent of blooming flowers across the peaceful landscape."

Actual segmentation output:

[(12, 489, 19, 51)]

Actual text extraction output:

"The river flows gently through the valley, reflecting the golden sunset. Birds sing in the trees, while a cool breeze carries the scent of blooming flowers across the peaceful landscape QC"

UT-09 / UT-11-2

Input:

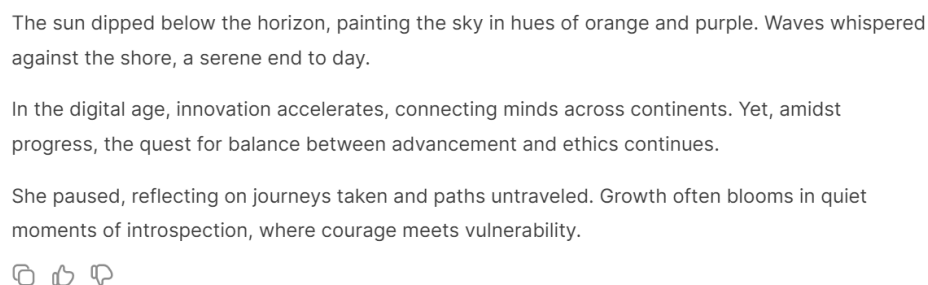


Figure 2. Multi-Region Text Image

Expected segmentation output (left, right, top, bottom):

[(14, 607, 15, 49), (14, 551, 73, 107), (14, 567, 126, 160)]

- Considered pass if each coordinate is within 5 pixels of the expected value

Expected text extraction output:

"The sun dipped below the horizon, painting the sky in hues of orange and purple. Waves whispered against the shore, a serene end to day. In the digital age, innovation accelerates, connecting minds across continents. Yet, amidst progress, the quest for balance between advancement and ethics continues. She paused, reflecting on journeys taken and paths untraveled. Growth often blooms in quiet moments of introspection, where courage meets vulnerability."

Actual segmentation output:

[(14, 479, 9, 39), (5, 446, 96, 124), (14, 431, 52, 81), (5, 431, 53, 81), (4, 447, 96, 125)]

Actual text extraction output:

"The sun dipped below the horizon, painting the sky in hues of orange and purple. Waves whispered against the shore, a serene end to day. In the digital age, innovation accelerates, connecting minds across continents. Yet, amidst progress, the quest for balance between advancement and ethics continues. She paused, reflecting on journeys taken and paths untraveled. Growth often blooms in quiet moments of introspection, where courage meets vulnerability."

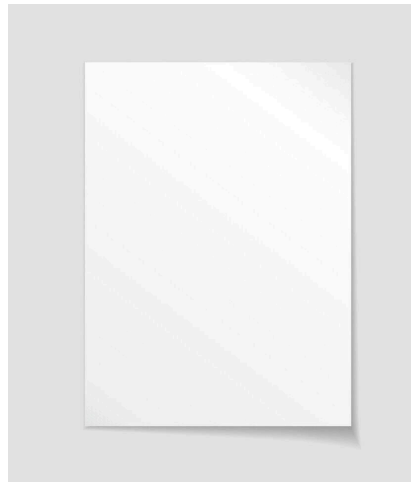
UT-10 / UT-12**Input:**

Figure 3. No Text Image

Expected segmentation output: []

Expected text extraction output: ""

Actual segmentation output: []

Actual text extraction output: ""

UT-13 to UT-17

For these 5 cases, $Lux=2.5 \times 2^{EV}$ is used to justify the brightness of images, where EV is the Exposure Value.

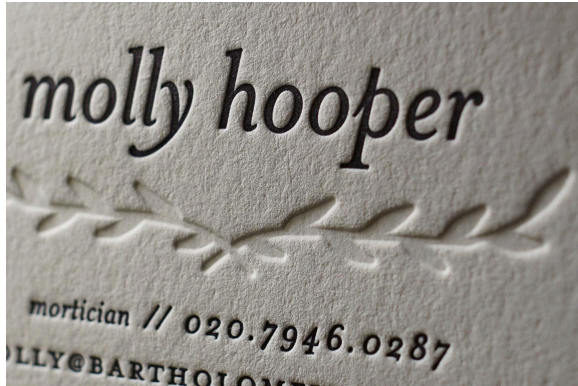


Figure 4. Normal Exposure Text Image

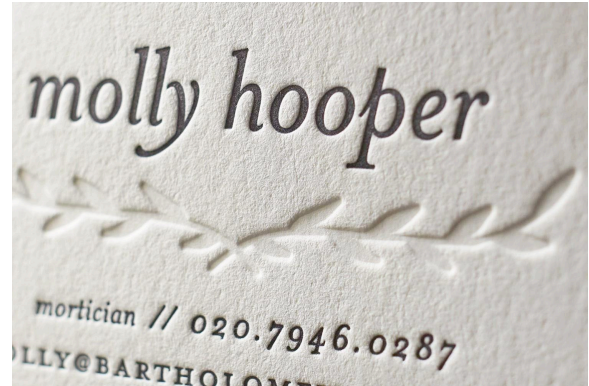


Figure 6. High Exposure Text Image

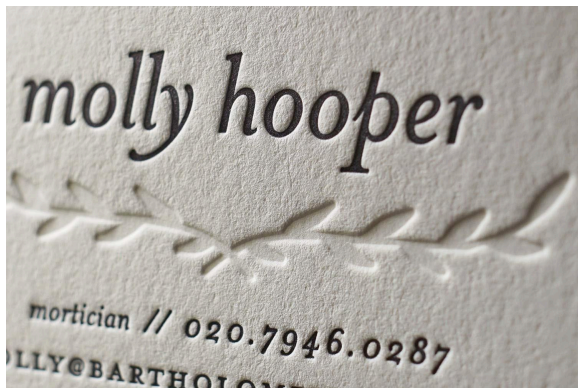


Figure 5. Under Exposure Text Image

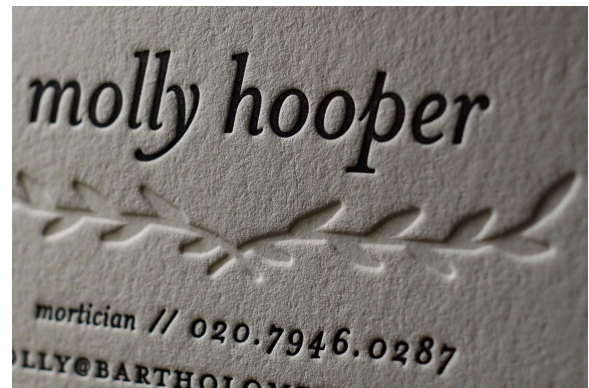


Figure 7. Under-exposure Text Image

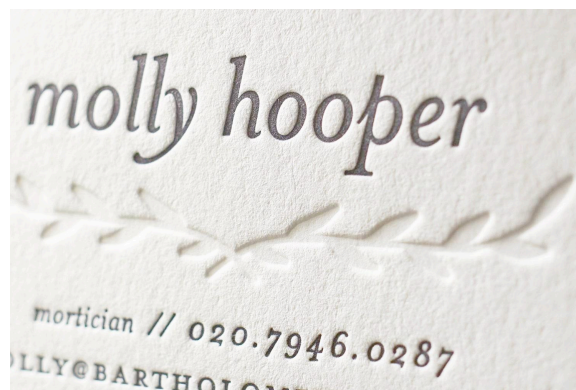


Figure 8. Over-exposure Text Image

Expected text extraction output: "molly hooper mortician//020.7946.0287 LLY@BARTH"

UT-10 Actual text extraction output: "molly hooper mortician//020.7946.028 LLYOBARTH"

UT-11 Actual text extraction output: "molly hooper mortician//020.7946.028 LLYOBARTH"

UT-12 Actual text extraction output: "molly hooper mortician//020.7946.028 LLYOBARTH"

UT-13 Actual text extraction output: "molly hooper mortician//020.7946.028 LLYOBARTH"

UT-14 Actual text extraction output: "molly hooper mortician//020.7946.028 LLYOBARTH"

UT-18 / UT-20

Input:



Figure 9. Arial Text Image

Expected segmentation output:

[(17, 300, 8, 108)]

- Considered pass if each coordinate is within 5 pixels of the expected value

Expected text extraction output: "LIEG TICKETS .NL"

Actual segmentation output: [(7, 318, 8, 158)]

Actual text extraction output: "TICKETS EG .NI"

UT-19 / UT-21

Input:

Calibri
Calibri
Calibri

Figure 10. Calibri Text Image

Expected segmentation output:

[(42, 629, 256, 438), (42, 629, 509, 680), (42, 640, 12, 187)]

- Considered pass if each coordinate is within 5 pixels of the expected value

Expected text extraction output: "Calibri Calibri Calibri"

Actual segmentation output: [(42, 629, 256, 438), (56, 660, 509, 680), (45, 624, 12, 187)]

Actual text extraction output: "Calibri Calibri Calibri"

IT-04

Close Distance & Standard Brightness

Input



Figure 11. Med-dist Std-light Input

Output



Figure 12. Med-dist Std-light Input

Result

Correctly identifies and translates "GREEN TEA WITH HONEY" and "PURE LEAF" but missing the "Real BREWED ICED TEA" at the bottom.

Close Distance & Low Brightness

Input



Figure 13. Med-dist Std-light Input

Output



Figure 14. Med-dist Std-light Input

Result

Correctly identifies and translates "GREEN TEA WITH HONEY" and "PURE LEAF" but mistakenly groups them in the same section. Missing the "Real BREWED ICED TEA" at the bottom.

Close Distance & High Brightness

Input



Figure 15. Med-dist Std-light Input

Output

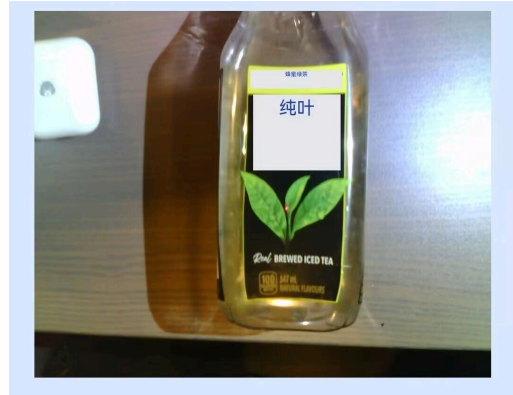


Figure 16. Med-dist Std-light Input

Result

Correctly identifies and translates "GREEN TEA WITH HONEY" and "PURE LEAF" but missing the "Real BREWED ICED TEA" at the bottom.

Medium Distance & Standard Brightness

Input

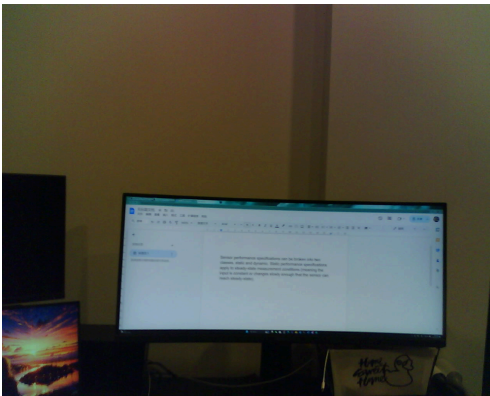


Figure 17. Med-dist Std-light Input

Output

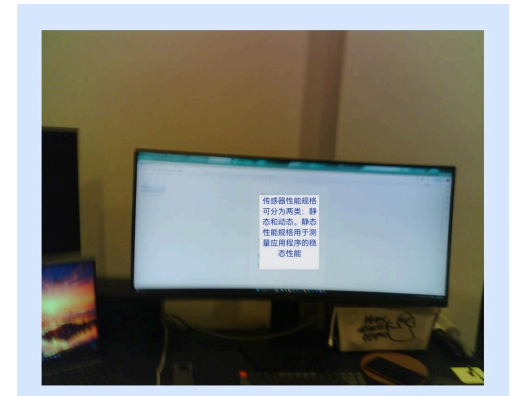


Figure 18. Med-dist Std-light Input

Result

Correctly identifies all text in the document

Long Distance & Standard Brightness

Input



Figure 19. Med-dist Std-light Input

Output

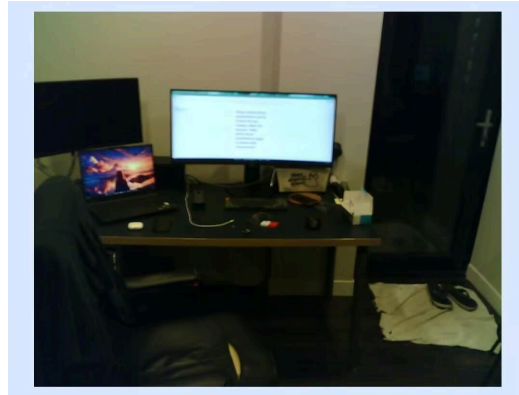


Figure 20. Med-dist Std-light Input

Result

Does not identify any text in the document

No Text & Standard Distance & Standard Brightness

Input



Figure 21. Med-dist Std-light Input

Output

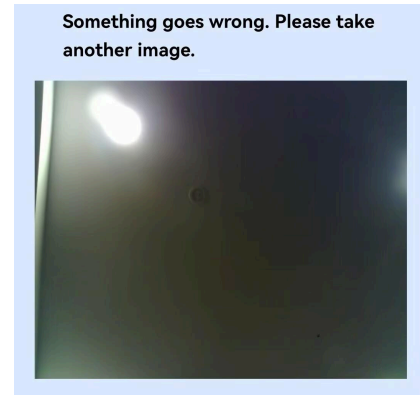


Figure 22. Med-dist Std-light Input

Result

Warning correctly displayed